

## 112 - AI-Based Analysis and Inconsistency Detection in Technical Drawings

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Allowed groups: 3 groups

### Project background & goals

In construction and modular building projects, multiple technical drawings are produced by different disciplines, including architectural, structural, and services engineers. These drawings may exist as 2D plans or as part of a BIM model. Because they are developed separately, inconsistencies and clashes often remain undetected until late in the design or construction phase. This leads to costly rework, delays, and coordination issues.

Reviewing and cross-checking drawings is currently a manual and time-consuming task that relies heavily on expert interpretation. With recent advances in artificial intelligence, particularly in computer vision and multimodal models, there is an opportunity to automate parts of this process by summarizing drawing content, explaining technical elements, and identifying inconsistencies across drawings.

The goal of this project is to develop an AI-based system that can analyze technical drawings, generate structured summaries and explanations of their content, and detect potential inconsistencies between different drawing types.

Specifically, the project aims to:

- Develop a model capable of interpreting information from 2D technical drawings or BIM exports.
- Generate clear, structured summaries of drawing content, such as key components, dimensions, and system elements.
- Compare multiple drawings and identify potential conflicts or inconsistencies.
- Evaluate the accuracy and reliability of the system using selected case examples.

### Project scope

This project involves developing a prototype AI system that can analyse selected 2D technical drawings or BIM exports, generate structured summaries explaining key elements, and detect inconsistencies between drawings from different disciplines. The work will focus on model development, comparison logic, and evaluation using sample drawing sets, without full integration into commercial BIM platforms.

### Project requirements

The system must be able to ingest selected 2D technical drawings or BIM model exports and extract relevant visual and textual information. It should generate structured summaries that explain key elements such as components, dimensions, labels, and system layouts. The system must also compare drawings from different disciplines and identify potential inconsistencies, such as missing elements, dimensional mismatches, or alignment conflicts. Outputs should be presented in a clear, structured report format highlighting detected issues and explanatory reasoning. The prototype should be modular, reproducible, and evaluated against sample drawing sets to assess accuracy and reliability.

### Required knowledge and skills

Students should have strong programming skills, preferably in Python, and a solid understanding of machine learning fundamentals. Experience with computer vision, multimodal AI models, or natural language processing will be highly beneficial. Familiarity with handling image data, structured outputs such as JSON, and basic software system design is expected. Knowledge of BIM concepts, CAD drawings, or engineering design documentation would be advantageous but not mandatory.

### **Expected outcomes and deliverables**

The project will deliver a working prototype capable of analyzing technical drawings, generating structured summaries, and detecting inconsistencies between selected drawing sets. Deliverables will include well-documented source code, a technical report describing the system design and evaluation, and a brief user guide explaining how to run the system and interpret the outputs.

### **Disciplines related to the project**

Software Development;Artificial Intelligence (Machine/Deep Learning, NLP);Computer Vision

### **Resources provided**

Students will be provided with sample 2D technical drawings and selected BIM model exports for development and testing. Access to approved AI model APIs and relevant computational resources will be arranged where required. Guidance on BIM concepts and drawing interpretation, along with regular technical supervision, will also be provided.